

## PROFESSIONAL EXPERIENCE

- 📅 2025.02 — Present **Department of Civil & Environmental Engineering, UIUC**  
Postdoc Research Associate (PI: Xiaojia Shelly Zhang)
- 📅 2024.05 — 2025.02 **Department of Mechanical Engineering, Tsinghua University**  
Postdoc Research Assistant, *Shuimu Tsinghua Scholar* (PI: Peng Wen)
- 📅 2018.09 — 2024.06 **School of Mechanical Engineering, Shanghai Jiao Tong University**  
Integrated MSc/PhD program in ME (Advisor: Ping Zhu) GPA: 3.69 Rank: 3/108  
**Thesis:** Energy-absorbing mechanism and design method of three-dimensional chiral mechanical metamaterials
- 📅 2014.09 — 2018.06 **School of Mechanical Engineering, Shanghai Jiao Tong University**  
Bachelor in Mechanical Engineering GPA: 3.95 Rank: 1/27

## RESEARCH PUBLICATIONS

## In Progress (Under Review)

- 1 **W.Y. Xu**, D. Hong, Z. Zhao, R.D. Kundu, and X.S. Zhang. **Impedance-Guided Programmable Transmission of Localized Deformation in Modular Soft Metamaterials**, preprint arXiv:2607.08966.
- 2 **W.Y. Xu**, and J.M. Liu. **Growth-Inspired Graph Generation and Inverse Design of Mechanical Lattices via Dot Matrices Database Augmentation and GCNN**, preprint arXiv:2609.29024.
- 3 Y.B. Wang\*, A.F. Ahmed\*, W.C. Li, **W.Y. Xu**, and X.S. Zhang. **Freeform Programmable Liquid Crystal Elastomers via Monolithic Integration of Co-Optimization and 4D-Printing**, 2026.
- 4 X.Y. Tai, **W.Y. Xu**, and P. Zhu. **Gradient metamaterials with moment-of-inertia-induced low-frequency vibration attenuation**, 2026.

## Journal Articles

- 1 H.L. Pang\*, **W.Y. Xu\***, J.G. Liu\*, H.J. Xu, Z.G. Wang, F. Song, Y. Tian, and P. Wen. **Rare earth redistribution governs competitive surface oxidation and grain refinement equilibrium in magnesium alloys**, *J. Magnes. Alloy*, vol. 22, p. 102231, 2026.
- 2 **W.Y. Xu\***, H.L. Pang\*, H.J. Xu, J.G. Liu, Y.F. Zheng, and P. Wen. **Process-driven microstructure design of 3D-printed porous magnesium alloy scaffolds with tunable biodegradation kinetics**, *Int. J. Mach. Tools Manuf.*, vol. 215, p. 104362, 2025.
- 3 J.M. Liu, B. Peng, **W.Y. Xu**, Y. Wei, and P. Wen. **Highly Efficient Discovery of 3D Mechanical Metamaterials via Monte Carlo Tree Search**, *Adv. Sci.*, vol. 12, p. e13771, 2025.
- 4 Z.Z. Song, **W.Y. Xu**, M.A. Valdebenito, and M.G.R. Faes. **Efficient forward and inverse uncertainty quantification for dynamical systems based on dimension reduction and Kriging surrogate modeling in functional space**, *Mech. Syst. Signal Pr.*, vol. 235, p. 112898, 2025.
- 5 Z.J. Pei\*, H.J. Xu\*, M.Z. Guo\*, **W.Y. Xu**, Y. Wen, F.P. Sun, T.Y. Zhang, B. Peng, P.Q. Zhao, L.K. Huang, M.Y. Wang, Z.S. He, J.Z. Liu, Z.C. Yang, Z. Zhang, P. Wen, and L.Y. Wen. **A soft-hard hybrid scaffold for osteochondral regeneration through integration of composite hydrogel and biodegradable magnesium**, *Biomaterials*, vol. 324, p. 123493, 2025.
- 6 **W.Y. Xu**, C. Zhou, H.Y. Zhang, Z. Liu, and P. Zhu. **A flexible design framework for lattice-based chiral mechanical metamaterials considering dynamic energy absorption**, *Thin-Walled Struct.*, vol. 203, p. 112108, 2024.
- 7 **W.Y. Xu**, H.Y. Zhang, Z. Liu, and P. Zhu. **Aperiodic design framework of chiral mechanical meta-materials considering crashworthiness**, *J. Mech. Eng.*, JME2023-1296, 2024. (In Chinese)
- 8 L. Zhang, **W.Y. Xu**, R.Y. Qiu, D.K. Xu, H.Y. Zhang, and P. Zhu. **Multiscale-based multiaxial fatigue model of short fiber reinforced polymer composites under high-cycle proportional loading**, *Compos. Part B-Eng.*, vol. 275, p. 111308, 2024.

- 9 W.Y. Xu, L. Zhang, B.Q. Zhang, H.Y. Zhang, Z. Liu, and P. Zhu. **Crushing behavior of contact-aided AlSi10Mg sandwich structure based on chiral mechanical metamaterials**, *Int. J. Mech. Sci.*, vol. 260, p. 108636, 2023.
- 10 W.Y. Xu\*, L.W. Wang\*, Z. Liu, and P. Zhu. **General assembly rules for metamaterials with scalable twist effects**, *Int. J. Mech. Sci.*, vol. 259, p. 108579, 2023.
- 11 W.Y. Xu, Z. Liu, L.W. Wang, and P. Zhu. **3D chiral metamaterial modular design with highly-tunable tension-twisting properties**, *Mater. Today Commun.*, vol. 30, p. 103006, 2022.

### Conference Proceedings/Oral Presentations

- 1 W.Y. Xu, B. Peng, and P. Wen. **SD-DAL: Structure discovery of elastic metamaterials via deep active learning**, *APS Global Physics Summit 2025*, Anaheim, CA, USA, Mar. 17-21, 2025.
- 2 W.Y. Xu, X.Y. Tai, and P. Zhu. **Bio-inspired Mechanical Metamaterials with Enhanced Energy Absorption**, *APS Global Physics Summit 2025*, Anaheim, CA, USA, Mar. 17-21, 2025.
- 3 W.Y. Xu, H.J. Xu, H.L. Pang, J.G. Liu, and P. Wen. **Corrosion resistance and mechanical properties of biodegradable WE43 magnesium alloy porous structure controlled by 3D printing layer thickness**, *The 2nd Ganjiang Academic Forum on Biomaterials and Medical Devices*, Ganzhou, China, Nov. 22-24, 2024.
- 4 W.Y. Xu. **High temperature oxidation treatment of magnesium alloys in additive manufacturing**, *2024 Chinese Society for Biomaterials (CSBM) Annual Symposium*, Weihai, China, Oct. 11-13, 2024.
- 5 W.Y. Xu. **Energy absorption mechanism of three-dimensional chiral mechanical metamaterials and their application in the frame of electric heavy-duty trucks**, *The 2nd Doctoral Academic Forum of the China Society of Automotive Engineers*, Beijing, China, Jun. 15-16, 2024.
- 6 W.Y. Xu, W.J. Wang, and P. Zhu. **GNN-based inverse design of three-dimensional aperiodic metamaterials enabling programmable shapes**, *APS March Meeting 2024*, Minneapolis, MN, USA, Mar. 3-8, 2024.
- 7 W.Y. Xu, H.Y. Zhang, Z. Liu, and P. Zhu. **On the crashworthiness of aperiodic chiral mechanical metamaterials: design and modeling method** in *J. Phys. Conf. Ser.*, vol. 2639, p. 012029, 2023.

### Patents

- 1 W.Y. Xu, P. Zhu, Z. Liu, and Y.F. Li. **Chiral mechanical metamaterial sandwich structures with size-effect-free twist and the applications**, China Patent CN115691719B, Jul. 29, 2025. (Authorized)
- 2 P. Zhu, L. Zhang, Z. Liu, W.Y. Xu, and Z.Z. Song. **Stiffness-based mixed rapid prediction method for fatigue life of SFRP**, China Patent CN116305990B, Jul. 22, 2025. (Authorized)
- 3 P. Zhu, W.Y. Xu, Z. Liu, L.W. Wang, and L. Zhang. **Automatic simulation system and method for strut-based metamaterial under multiple working conditions**, China Patent CN114297877B, Dec. 3, 2024. (Authorized)
- 4 P. Zhu, W.Y. Xu, Z. Liu, and M.S. Li. **Mesoscopic structural optimization methods**, China Patent CN110362912B, Nov. 8, 2022. (Authorized)
- 5 W.Y. Xu, P. Zhu, and W.Z. Guo. **Foldable multi-form electric vehicle**, China Patent CN109178180B, May 5, 2020. (Authorized)
- 6 H.Y. Zhang, W.Y. Xu, Z. Liu, and P. Zhu. **Implementation method for aperiodic chiral mechanical metamaterial**, China Patent CN120781587A, Oct. 14, 2025. (Pending)
- 7 Z. Liu, W.Y. Xu, and P. Zhu. **Twist angle measurement fixture for compression-torsional testing of chiral mechanical metamaterial**, China Patent CN116026678A, Apr. 28, 2023. (Pending)

### Book and Chapter

- 1 P. Zhu. **Advanced design theory and methodology**. Beijing, China: China Machine Press, 2023, ISBN: 978-7-111-71470-5. (in Chinese) (Chapter 3 and 7)

MISCELLANEOUS EXPERIENCE

Representative Research Funding Acquisition

- National Natural Science Foundation of China (Grant No. 12472119)** 2025.01 — 2028.12  
Energy absorption mechanism and design of 3D chiral mechanical metamaterials based on PINN
- Shanghai Natural Science Foundation (Grant No. 23ZR1431600)** 2023.04 — 2026.03  
Research on energy absorption mechanism and optimization design method of 3D chiral metamaterials
- Shanghai Natural Science Foundation (Grant No. 21ZR1431500)** 2021.04 — 2024.03  
Research on data-driven multi-scale optimization design method of mechanical metamaterial

Leadership & Service

- Student President of Graduate Union**, Shanghai Jiao Tong University 2019.05 — 2020.07
- Certified Volunteer**, 2025 Christie Clinic Illinois Race Weekend 2025.04
- Certified Volunteer**, 2016 Shanghai International Marathon 2016.11

Teaching Assistant

- Undergraduate Course *Open Source and Modeling* Spring Semesters, 2021-2022
- Undergraduate Course *Fundamentals of Manufacturing Processes* Fall Semesters, 2019-2023
- Undergraduate Course *The Way To Success* Fall 2015

Industry Experience

- Face Recognition Project Internship**, Honeywell (China) Ltd., Shanghai 2018.07 — 2018.08

AWARDS & HONORS / FELLOWSHIPS

- National Scholarship (1%)**, Ministry of Education of P. R. China 2017/2019/2023
- National Inspirational Scholarship**, Ministry of Education of P. R. China 2016
- Merit Student Award**, Shanghai Jiao Tong University 2017
- Outstanding Graduate Award**, Shanghai Jiao Tong University 2018
- Excellent Graduation Design Award**, School of Mechanical Engineering, SJTU 2018
- 1<sup>st</sup> Place in Road Test Competition of PACE**, General Motors (North America) Ltd. 2018
- 2<sup>nd</sup> Place of 30<sup>th</sup> International Design Contest ROBOCON**, MIT 2019
- Inspirational Individual Award**, Shanghai Jiao Tong University 2020
- Excellence Teaching Assistant Award**, Shanghai Jiao Tong University 2020
- Shanghai PhD Outstanding Graduate Award**, Shanghai Jiao Tong University 2024
- APS DMP Ovshinsky Travel Award**, APS Division of Materials Physics 2024
- APS FGSA Graduate Research Excellence Travel Award**, APS FGSA 2024
- Class D Municipal High-level Talent**, Ganzhou, Jiangxi Province, P. R. China 2024
- Shuimu Tsinghua Scholar Fellowship**, Tsinghua University 2024
- Excellent PhD Dissertation**, School of Mechanical Engineering, SJTU 2025
- Scholarship and Teaching for Engineering Postdocs (STEP) Program**, GCOE, UIUC 2025
- Nominated Full Membership**, Sigma Xi, The Scientific Research Honor Society 2026

SKILLS

- Languages** Strong reading, writing, and speaking competencies for English and Mandarin Chinese.
- Coding** Java, Python, C/C++, SQL (Postgres), JavaScript, MATLAB, Latex.
- CAD** Solidworks, Unigraphics NX, AutoCAD, Blender, nTopology, ParaView
- CAE** ABAQUS, ANSYS, COMSOL Multiphysics, LS-DYNA, Hypermesh, FEniCSx
- Hardware** Arduino, STM8/STM32, Raspberry Pi
- Manufacturing** CNC, Casting, Additive Manufacturing (FDM, SLA, SLM, SLS), DIY 3D-Printer